

Can BERT Understand How You Feel?

Sentiment Analysis & Text Classification using a Pre-trained BERT Model



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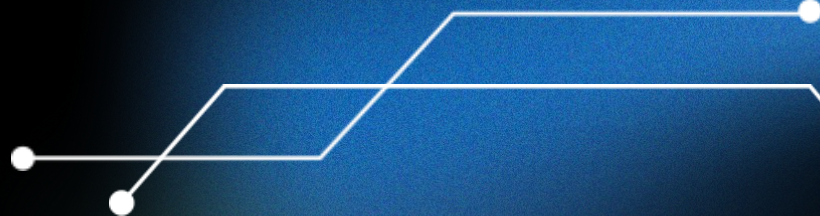
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The Words Aren't Enough

Context changes the meaning

Sentiment Analysis : *The task of identifying the emotional tone or sentiment expressed in text.*



Same word – **SICK**
But entirely different
meanings that is because
sentiment depends on
context



So how does the machine figure out what
we actually mean?



BERT uses context

BERT : A Powerful Language Model That Understands Context.



PRE TRAINED BERT

Learns general language representation from large text corpora.

Model : bert-base-uncased



FINE TUNING

Adapts BERT to our classification task using the SMILE dataset



7 CLASS CLASSIFIER

Predicts the emotion of the tweet

Output Classes

The model classifies each tweet into one of 7 emotions:



happy



angry



sad



surprise



disgust



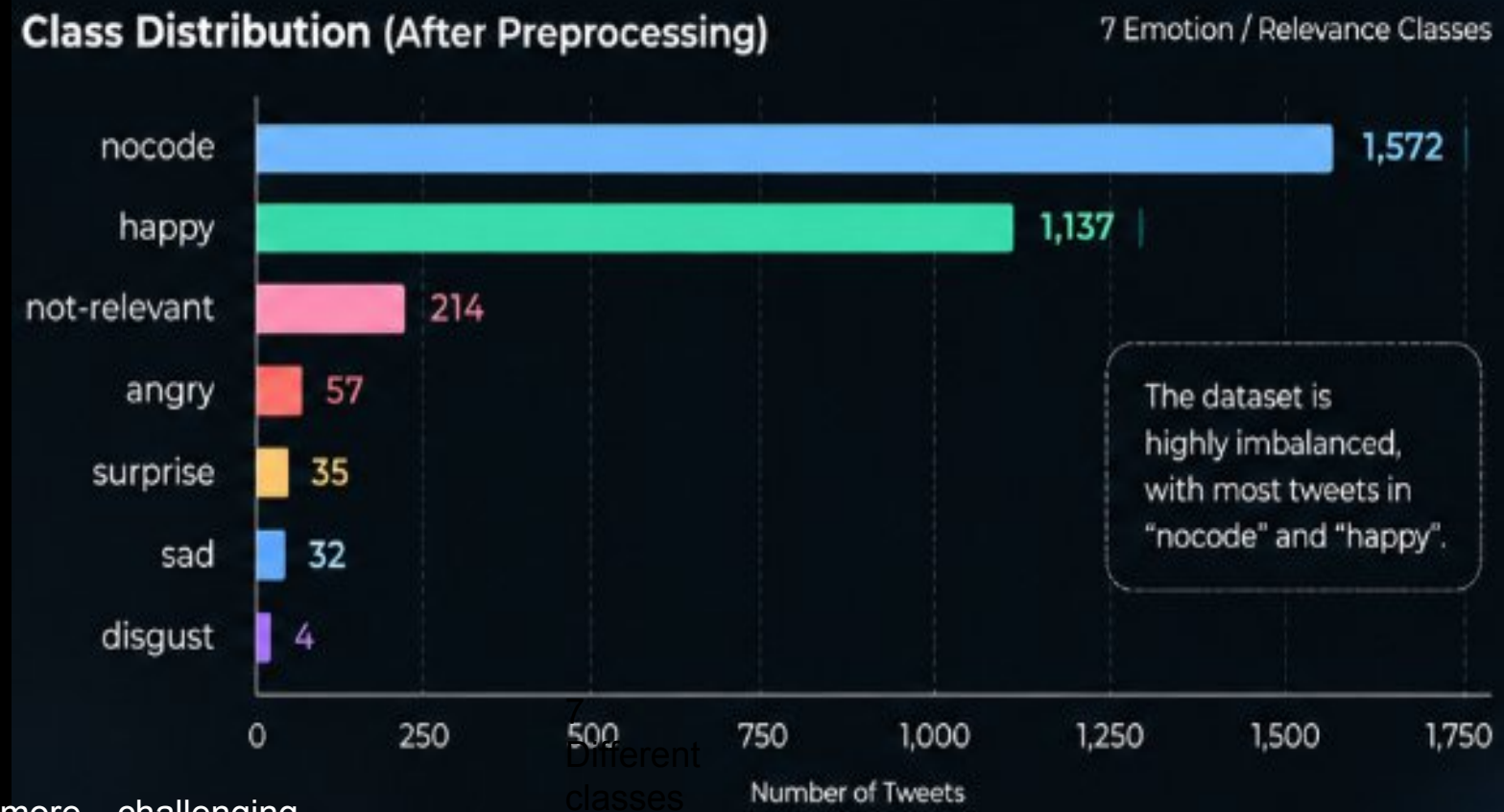
no code



not-relevant

The Data

Real tweets that propose real complexity.



This imbalance makes the classification task more challenging.
Especially for the emotions that have the least data entries.

From Text To Prediction

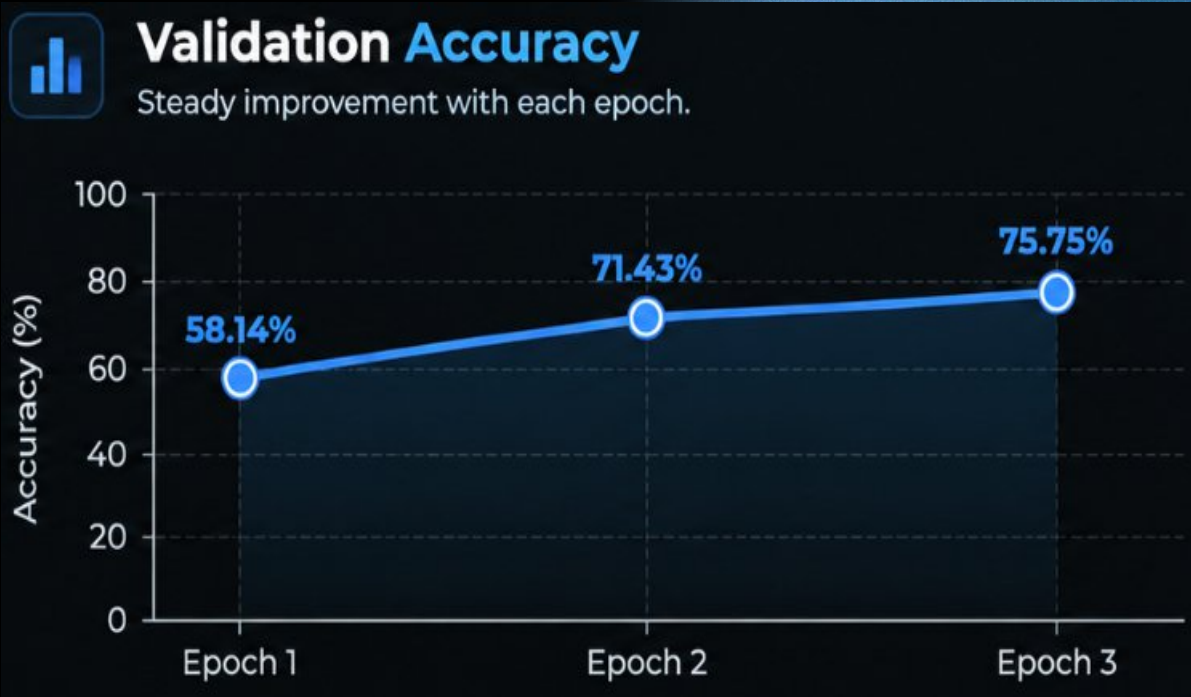
How the BERT model actually works



A raw tweet goes into the model, a prediction comes out

How The Model Learns

Validation accuracy rises while training loss falls across 3 epochs



Accuracy ↑

Loss ↓

The Results

How well did BERT perform on unseen data?



76.16 %

TEST ACCURACY

Overall correctness on
unseen data



54.75 %

MACRO-F1

Equal Weight to every class
Reveals performance on minority
classes



76.93 %

WEIGHTED-F1

Influenced by majority classes
Higher due to class imbalance in the
dataset

Accuracy alone cannot be used to judge the model performance..
The F1 scores clearly addresses the underlying problem of an imbalanced
dataset

So.. Can BERT Actually Understand What You Feel?

BERT doesn't really feel emotions, it learns patterns and context in language to classify what the text expresses



01

CONTEXT MATTERS

BERT uses surrounding words to understand meaning.



02

DATA MATTERS

Class imbalance can make overall accuracy misleading.

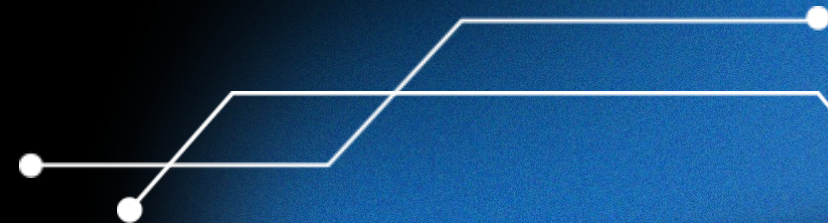
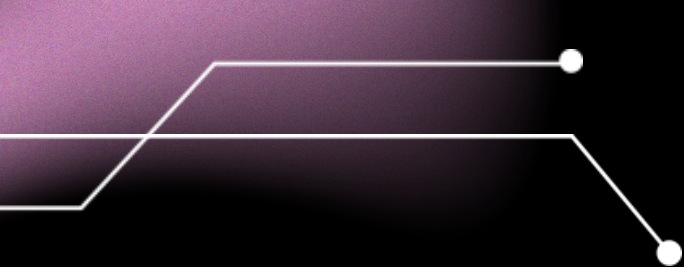


03

EVALUATION MATTERS

Macro-F1 gives a fairer view across all classes, especially minority ones.

Understanding sentiment isn't about reading emotions — it's about learning the patterns that express them



References & Materials

01 — BERT

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K.

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding.

02 — Sentiment Classification using BERT

GeeksforGeeks

Sentiment Classification Using BERT

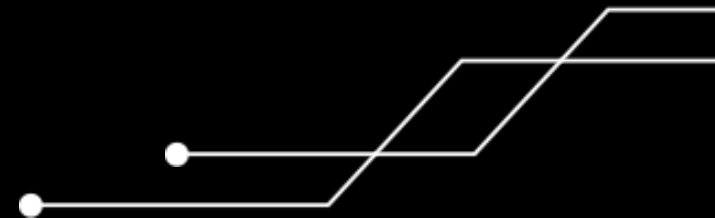
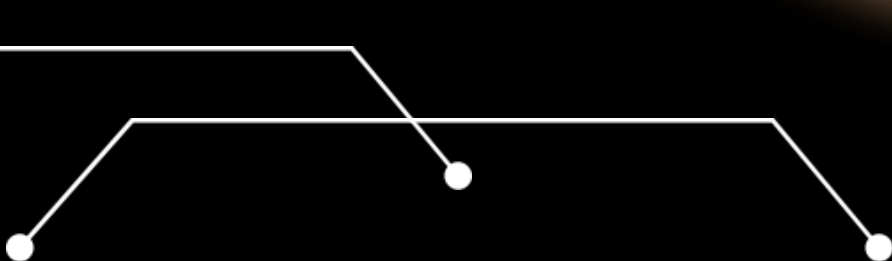
03 — Dataset

SMILE Twitter Emotion Dataset

Used for emotion/sentiment classification.

04 — Implementation

Python · PyTorch · Hugging Face Transformers · Google Colab/Jupyter Notebook



Explore The Project

Code. Dataset. Notebook. Presentation.

